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CASE STUDY

CHURHAT BYPASS TUNNEL (REWA - SIDHI)

INDUSTRY	Infrastructure
CUSTOMER	Dilip Buildcon Limited
OBJECTIVE	Ensure safety, efficiency, and remote control of tunnel operations
LOCATION	Rewa-Sidhi section of NH 75E in the state of Madhya Pradesh
AUTHORITY	NHAI
PROJECT	2.95 + 2.65 KM Long Main Tunnel
	6 Lane Road
	7 Cross Passages

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CHALLENGES

- **Environmental Damage:** High humidity and chemicals harm equipment
- **Electromagnetic Interference:** EMI affects electrical systems reliability
- **Lighting Requirements:** Specialized lighting needed for harsh conditions
- **Communication Issues:** Reliable communication infrastructure is challenging
- **Sensor Integration:** Integrating sensors and detectors is complex
- **Emergency Design:** Systems must support emergency response and evacuation
- **Environmental Constraints:** Systems operate within temperature, humidity ranges





- **Water Protection:** Systems must withstand water ingress, floods
- **Cable Laying:** Confined space, visibility issues complicate cable laying
- **Distance Laying:** Long-distance cable laying risks signal degradation
- **Signal Attenuation:** Long cable runs cause signal loss
- **Geometry Complexity:** Tunnel geometry complicates cable tray installation
- **Cable Tray Support:** Providing support for trays over distances
- **Coordination:** Coordinating cable installation with other activities

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SOLUTION

- **Protective Enclosures:** Use IP67 enclosures for equipment protection
- **Redundant Power:** Implement redundant power, UPS, backup generators
- **Surge Protection:** Install surge protection for voltage spikes
- **Reliable Communication:** Use fiber optic, wireless networks for communication
- **Automation Redundancy:** Implement redundant systems for continuous operation
- **Fault-Tolerant Design:** Design systems with fault-tolerant architecture

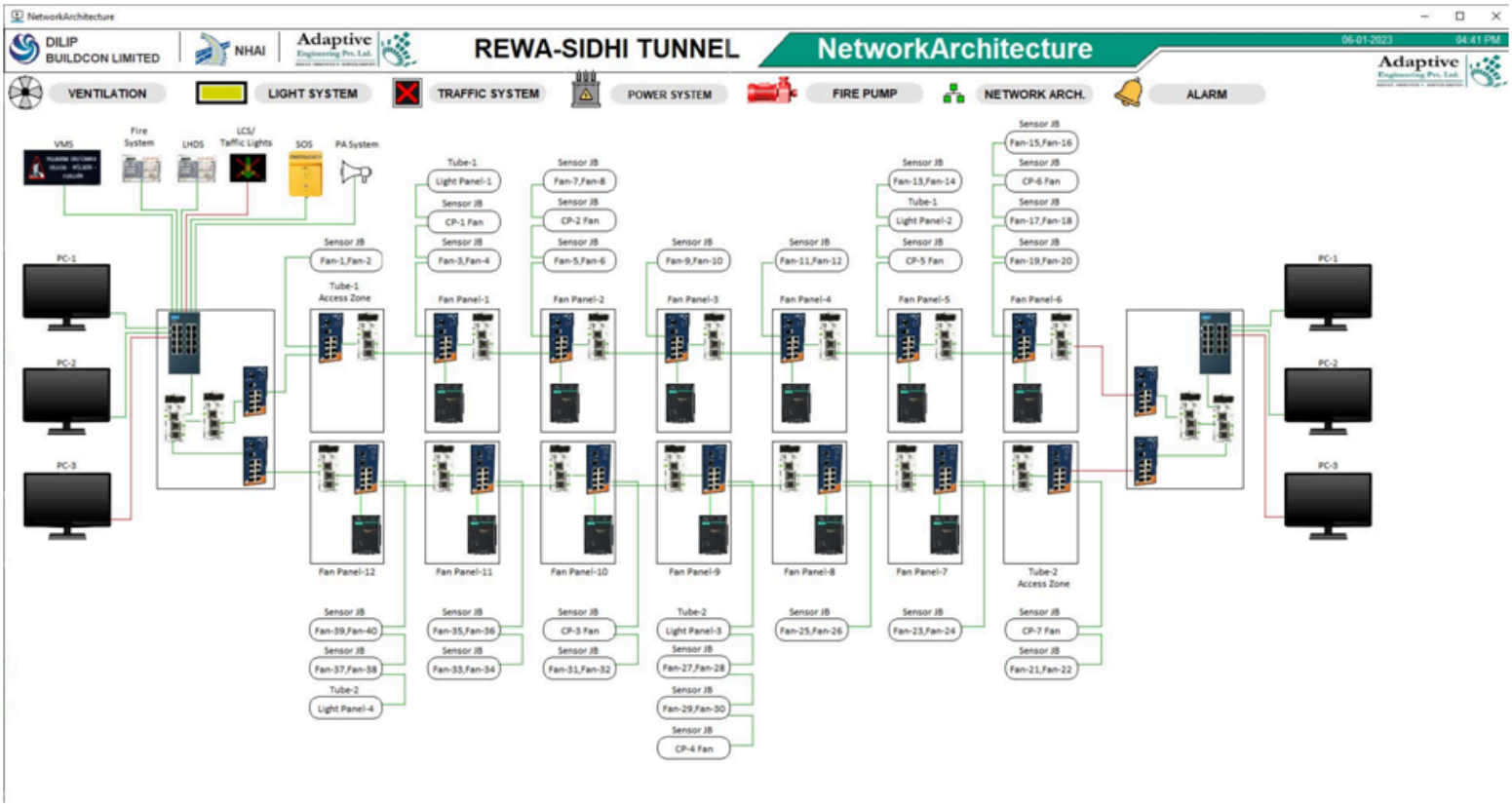




- **Remote Monitoring:** Implement real-time remote monitoring, control systems
- **Fire-Resistant Cables:** Use fire-resistant cables to prevent fire spread
- **Cable Management:** Implement systems to organize and secure cables
- **Custom Cable Trays:** Design trays tailored to tunnel's geometry
- **Support Systems:** Use brackets, hangers for cable tray stability

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ARCHITECTURE





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VALUE CREATION

- **Improved Safety:** Ensures secure and safe tunnel operations
- **Increased Efficiency:** Streamlines operations, enhancing overall performance
- **Energy Savings:** Reduces energy consumption through efficient systems
- **Reduced Labor Costs:** Minimizes manual labor requirements and costs
- **Enhanced Traffic Flow:** Optimizes traffic management for smoother flow





- **Faster Incident Response:** Enables quick detection and response to incidents
- **Improved Maintenance Scheduling:** Facilitates timely and efficient maintenance
- **Reduced Downtime:** Minimizes system downtime, ensuring operational continuity
- **Increased Tunnel Uptime:** Maximizes the tunnel's operational availability

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